One daily link row. Assignment totals preserved.

A new postprocessing export combines each directed link's incremental speeds and approximate interval volumes across AM, MD, PM and NT. S3 supplies the within-period pattern; one multiplier per link and period preserves the assigned total.

Regional links	Five-minute speeds	Five-minute volumes	Maximum total error
33,350	288	288	6.55e-11 vehicles

Step-by-step implementation

1. Resolve outputs and scopes. Run after both combined performance files are created in the resolved DTALite folder. Use exactly their jurisdiction and regional roadway scopes.

2. Identify the same directed link. Join converted network attributes by from_node_id + to_node_id. Export the network link_id, keeping the assignment ID separately in the audit.

3. Build the complete time grid. Assign each timestamp to one configured period, including NT across midnight. Preserve supplied speeds and identify any missing timestamps explicitly.

4. Resolve link-level S3 inputs. Read known free speed, capacity and lanes. Derive critical speed or m from one another; apply a documented prior only where the information is missing.

5. Convert and conserve. Calculate whole-link flow, multiply by interval hours, sum within each period, then apply its assignment-total ratio to every interval.

6. Write and audit. Write one row per unique directed link. Save parameter provenance, effective period attributes, scaling factors, exceptions and conservation errors separately.

Implementation location

Branch: **codex/s3-speed-volume-export**
Calculation: postprocessing/pipeline/speed_volume.py
Automatic trigger: postprocessing/runner.py
Regression tests: tests/test_speed_volume.py
Reproduction: scripts/validate_speed_volume.py

Input files, columns and units

Inputs are the existing period link.csv and link_performance.csv files. The processed combined DataFrame supplies period-owned performance values; the exporter rereads converted link.csv for network attributes. Positive finite values are selected row by row.

Quantity	Input columns / precedence	Interpretation
Identity	Both files: from_node_id, to_node_id	Normalized directed pair. Opposite directions remain distinct; duplicate/unmatched pairs fail.
Output ID / type	link.csv: link_id, link_type	First configured period containing that link. Internal performance IDs stay in the audit.
Assigned total V	link_performance.csv: volume	Whole-link assignment count for that period. Never divide by lanes.
Incremental speeds	spd_mph_HH:MM; legacy spd_HH:MM or spd_kmph_HH:MM	Use mph; convert km/h with 1.609344. Prefer explicit mph for duplicate representations.
Free speed vf	link.csv:vdf_free_speed_mph; performance:free_speed_mph; link.csv:free_speed	First two are mph. Generic link.csv free_speed is km/h and is divided by 1.609344.
Lane count	link.csv:lanes	Directed lane count for the current period. Assume 1 only when unavailable/invalid; audit this.
Capacity C	performance:link_capacity; link.csv:capacity; performance:lane_capacity	link_capacity is already veh/hour/link. Other two are veh/hour/lane; multiply by current lanes once.
Critical speed vc	speed_at_capacity_mph, vc_mph, fd_vc_mph	Performance fields first, then network equivalents. Require 0 < vc < vf.
Shape m	s3_shape_m, s3_m	Performance first, then network. Derive missing m from vf and vc.

Whole-link example

A three-lane link with capacity 1,800 veh/hour/lane has C = 5,400 veh/hour. If link_capacity already equals 5,400, use 5,400 directly. The resulting density, flow and interval volumes all describe the whole directed link.

QVDF cutoff_speed_mph, vdf_s, vdf_n and boundary speed anchors are not S3 speed-at-capacity or S3 shape parameters.

S3 inversion and period scaling

Derive parameters before assuming them

If vf and m are known, derive vc. If vf and vc are known, derive m. When vc and m conflict, supplied valid vc is authoritative and m is recomputed. If vf is missing but vc and m exist, derive vf first.

```
vc = vf / 2^(2/m)
m = 2 ln(2) / ln(vf/vc)
kc = C / vc
```

When neither critical speed nor shape is supplied, use CBI's vc = 0.70 vf, giving m = 3.887. Known network vf is preserved, including urban speeds below 35 mph. Missing vf falls back to speed p95 clipped to 35-85 mph; absent data uses 35 mph. Missing C uses the CBI batch prior of 1,800 veh/hour/lane times lanes.

Invert each interval's speed

```
v* = min(v, 0.99 vf)
r = max(vf / max(v*, 1 mph), 1.00001)
k = kc [max(r^(m/2) - 1, 1e-8)]^(1/m)
q = min(v* k, C) [veh/hour/link]
u_i = q_i dt_i_hours [vehicles/link]
```

These numerical safeguards follow CBI's inverse S3 function. The code evaluates density in log form to avoid exponent overflow. The saved supplied speeds are preserved; the 0.99 vf limit applies inside inversion.

One common ratio for all intervals of one period

```
U = sum(u_i over the period)
R = assigned period volume V / U
approx_volume_i = R u_i
sum(approx_volume_i) = V
```

AM, MD, PM and NT each get their own R for each directed link. Floating-point residue is corrected in the largest interval; no rounding to integer vehicles is applied.

Illustrative five-minute bin	Raw interval count	R	Saved count
1	100	2.0	200
2	200	2.0	400
3	150	2.0	300
Period total	450	V / U = 900 / 450	900 = assignment

The CBI reconstruction module uses an additive capacity-constrained adjustment. This implementation uses the requested multiplicative adjustment. Final scaled flow may exceed C; the audit records it rather than clipping away assigned demand.

File schema and explicit assumptions

New file beside combined outputs	Contents
link_speed_volume.csv	Jurisdiction-scoped daily rows
link_speed_volume_regional.csv	Regional daily rows
Corresponding *_audit.csv files	One record per link and period; parameters, sources, totals, scale and flags

Daily column order

1. link_id, from_node_id, to_node_id, link_type, free_speed (mph), capacity (veh/hour/link), lanes.
2. All spd_mph_HH:MM columns in clock order.
3. All approx_volume_HH:MM columns in the same clock order. Values are fractional vehicles per interval.

The seven static fields describe the first configured period containing a link. Each calculation uses the actual attributes for its own period; the audit retains those effective values. A link absent from a period retains blank cells for that period.

Cases handled explicitly

Case	Behavior / audit
Missing overnight timestamps	Infer the regular grid. Fill absent columns with the same link's assigned period average speed_mph. Audit imputed_speed_intervals and imputation_source.
Supplied invalid speed cells	Keep the original cell. Give it zero raw flow weight and record invalid_speed_intervals. Other valid intervals receive the allocation.
Positive assignment, zero raw S3 total	Allocate in proportion to interval duration. Status uniform_duration_fallback; scale factor is blank.
Zero assigned volume	Every interval count is zero.
Duplicate/unmatched pairs, invalid assigned volume, overlapping periods	Clear error; no silent attribution of another link's parameters.
Legacy file with no speed columns	Skip new exports with a warning. Clear stale same-named exports; existing combined/statistics behavior remains.

Discovered in the existing true-shape run

NT is configured as 19:00-06:00, but the supplied NT speed columns cover only 19:00-23:55. The new export explicitly fills 00:00-05:55 (72 five-minute bins) with the link's assigned NT average speed. These are estimated speeds, not additional native model samples.

Validation on the true-shape network

Reprocessed copies of the existing DTALiteNetworkInput_TRUESHP_run outputs through the normal postprocessing entry point. This tests export/integration using completed assignment results; assignment itself was not rerun. All 103 automated project tests passed in a temporary checkout outside Dropbox.

Scope	Unique links	Columns	Speed / volume intervals
Jurisdiction	12,559	583	288 / 288
Regional	33,350	583	288 / 288

Every exported link/period total checked after reading the CSV

Scope	Period	Links	Bins	Assigned = exported*	Max error
Jurisdiction	AM	12,559	36	26,245,271.057	1.46e-11
Jurisdiction	MD	12,559	72	49,409,812.983	4.37e-11
Jurisdiction	PM	12,559	48	42,533,244.954	2.18e-11
Jurisdiction	NT	12,559	132	32,997,768.242	4.00e-11
Regional	AM	33,343	36	68,270,173.453	3.64e-12
Regional	MD	33,350	72	126,063,225.338	4.37e-11
Regional	PM	33,343	48	109,365,773.972	7.28e-12
Regional	NT	33,350	132	83,825,741.966	6.55e-11

*Equality within absolute 1e-8 and relative 1e-12 numerical tolerance. Error is the maximum per-link difference in vehicles after CSV round-trip, rather than only a network-wide comparison.

What the test confirmed

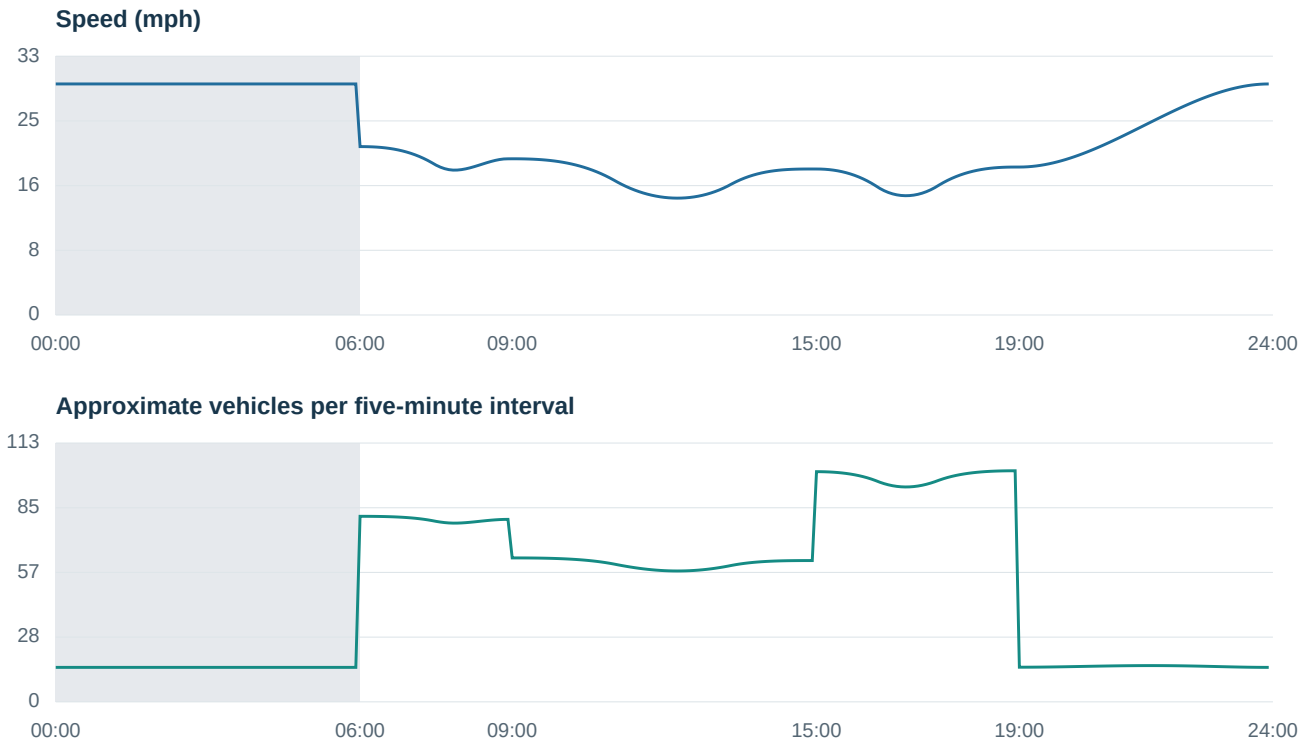
All 133,386 regional link/period records used their native link_capacity and network vdf_free_speed_mph. Critical speed was unavailable, so all used 0.70 vf and derived m. There were 0 invalid speed cells and no duration-allocation fallbacks. The regional audit records 2,401,200 filled overnight cells and 307,395 intervals above capacity after scaling.

931 regional links changed capacity between periods, confirming why period-specific attributes must be used. The raw input SHA256 hashes were unchanged: **True**.

Regression coverage includes S3 capacity recovery, lane conversion, independent scaling, midnight ownership, alternate speed units, missing timestamps, zero/invalid speeds, ID changes, absent-period links, duplicate-pair rejection and both automatic output scopes.

Example daily profile and reproducibility

Regional example: link 8250, directed pair 20000 to 20797. The shaded 00:00-06:00 segment uses the audited NT average-speed fill. Vertical levels in the volume chart reflect independent period scale factors.



Period	Assigned volume	Raw S3 volume	Scale R	Exported total
AM	2,870.499	1,453.731	1.97457	2,870.499
MD	4,348.883	2,792.527	1.55733	4,348.883
PM	4,715.883	1,858.639	2.53728	4,715.883
NT	2,013.428	5,295.633	0.38021	2,013.428

Reproduce and inspect

Run scripts/validate_speed_volume.py with --source pointing to existing period outputs and --output to a new test folder. It stages copies, runs normal postprocessing, writes validation_summary.json and ten-row sample CSVs, and hashes source inputs before and after.

Outputs for this report: **nvta-run/s3-speed-volume-validation-20261105-final/**
Specification: **docs/s3-speed-volume.md**
All requested changes are on **codex/s3-speed-volume-export**. Publishing is a separate release step after review.

Reference: CBI fundamental_diagram.py (pinned source); s3_speed, inverse_s3_flow and synthetic_s3_context. The 1,800 veh/hour/lane prior is from CBI batch.py. No CBI dependency was added to the installed workflow.